

Applications of tree data structure

Why Tree?

Unlike Array and Linked List, which are linear data structures, tree is hierarchical (or non-linear) data structure.

1. One reason to use trees might be because you want to store information that naturally forms a hierarchy. For example, the file system on a computer:
file system

```
      /    <-- root
     /      \
    ...      home
           /      \
        ugrad    course
         /      / |  \
        ...    cs101 cs112 cs113
```

2. If we organize keys in form of a tree (with some ordering e.g., BST), we can search for a given key in moderate time (quicker than Linked List and slower than arrays). [Self-balancing search trees](#) like [AVL](#) and [Red-Black trees](#) guarantee an upper bound of $O(\log n)$ for search.
3. We can insert/delete keys in moderate time (quicker than Arrays and slower than Unordered Linked Lists). [Self-balancing search trees](#) like [AVL](#) and [Red-Black trees](#) guarantee an upper bound of $O(\log n)$ for insertion/deletion.
4. Like Linked Lists and unlike Arrays, Pointer implementation of trees don't have an upper limit on number of nodes as nodes are linked using pointers.
- 5.

Other Applications :

1. Store hierarchical data, like folder structure, organization structure, XML/HTML data.
2. [Binary Search Tree](#) is a tree that allows fast search, insert, delete on a sorted data. It also allows finding closest item
3. [Heap](#) is a tree data structure which is implemented using arrays and used to implement priority queues.
4. [B-Tree](#) and [B+ Tree](#) : They are used to implement indexing in databases.
5. [Syntax Tree](#): Used in Compilers.
6. [K-D Tree](#): A space partitioning tree used to organize points in K dimensional space.
7. [Trie](#) : Used to implement dictionaries with prefix lookup.
8. [Suffix Tree](#) : For quick pattern searching in a fixed text.
9. [Spanning Trees](#) and shortest path trees are used in routers and bridges respectively in computer networks
10. As a workflow for compositing digital images for visual effects.